

Domain specific abductive attention

Gary Nan Tie, Oct 7, 2025

Abstract

A paradigm shift from LLMs as stochastic parrots to AI systems that use a mathematically grounded structure (fibrations and liftings) to perform constrained, explainable, abductive reasoning within specialized domains, directly addressing the critical flaws of hallucination and lack of domain specificity.

Introduction

Somewhat surprisingly the syntax of human language captures enough semantics through word association to enable large language models (LLM) like ChatGPT to work. Indeed attempts to mathematically model linguistics pale in comparison to what transformer attention mechanisms have done, like passing the Turing test. LLM however are not without flaws, so called hallucinations are problematic to fix, and going from generic association to domain specific specialization ad hoc.

LLM are typically trained on humongous swaths of internet data that contain facts, misinformation and disinformation alike. Suppose we wanted both a conversational interface and domain specific content, like medicine for example, so that doctors could readily consult with a LLM to make a medical diagnosis. What is not needed are random uninformed medical opinions or anecdotal evidence. An obvious solution is to take a general pre trained LLM and train it on a specialized medical database. The issue is transformer attention mechanisms which use scaled dot product are still nonetheless learning linguistic statistical correlations. However medical diagnosis requires abductive reasoning, hypothesizing potential diseases that could explain a patient's symptoms.

Attention Entanglement [1] on the other hand uses a fibration lifting to update both queries and keys to generate values that are explainable not just probable alone. Moreover, these liftings refine when their entropy is being lowered, that is, there is greater certainty in the explanations. As we will see such liftings can also be coherently specialized to domain specific knowledge to perform abduction.

Let's continue with the medical diagnosis example. Keywords in medicine determine a subset of values $V^\wedge$, of which, the inverse image under semantics $p: K \rightarrow V$, then defines a corresponding subset of keys $K^\wedge$. Fibration lifting $\sigma(p)$ then naturally defines a domain specific lifting $\sigma^\wedge$ when restricted to $V^\wedge$. For the medical domain, specialized lifting $\sigma^\wedge$ determines query and key updates to generate values specifically in answer to medical questions.

Upshot: one does not have to start from scratch all over again to perform domain specific abduction, rather one can adapt the general reasoning already learnt via fibration abductive attention on the corpus. This structured coherent form of transfer learning not only enables systematic discovery of new knowledge but also refinement of abductive reasoning, sharpened by the specifics of each domain of interest encountered; an ever growing virtuous spiral.

Fibrations are employed in type theory as categorical models of dependent types. In the context of transformer attention mechanisms we take a different tack, examining the logic of fibration liftings. A Joyal-Tierney symbolic calculus diagram gives a fibration whose lifting updates queries and keys in tandem. To address domain specificity, we introduce subject keywords to restrict attention. Generically, a subject is a subset of values, and relationships between them, can be specified by Venn diagram logic, in particular the three existential graphs [2] of Charles Sanders Peirce; alpha is isomorphic to propositional logic, beta is isomorphic to first-order logic, and gamma is (nearly) isomorphic to normal modal logic. This means detailed predicates on relationships between different subjects can be expressed, capturing domain specific knowledge.

Characterize a domain of interest by an existential graph predicate on certain relationships between relevant subjects, to determine a subset of values $V^\wedge$. A fibration lifting σ , determined by query update $m: Q \rightarrow Q'$ and semantics $p: K \rightarrow V$, learned on the corpus, then naturally defines a domain specific lifting $\sigma^\wedge$ when restricted to $V^\wedge$. Abductive inference learned across multiple disciplines, can be naturally specialized without retraining, to achieve customized attention, that coherently generates focused domain specific answers

Following the notation of [1] for Attention Entanglement:

Corpus input data

key-value pairs $(k_j, v_j) \in K \times V$, $j \in [m]$

query-update pairs $(q_i, q'_i) \in Q \times Q'$, $i \in [n]$

Learn NN $p: K \rightarrow V$ such that $p(k_j) \approx v_j$, $j \in [m]$

and NN $m: Q \rightarrow Q'$ such that $m(q_i) \approx q'_i$, $i \in [n]$

Given p and m , learn $\sigma: Q \otimes V \rightarrow Q' \otimes K$ such that:

$$\begin{array}{ccc}
 Q \otimes K & \xrightarrow{m \otimes 1} & Q' \otimes K \\
 \downarrow \text{1} \otimes p & \searrow \sigma & \downarrow \text{1} \otimes p \\
 Q \otimes V & \xrightarrow{m \otimes 1} & Q' \otimes V
 \end{array}
 \quad \text{commutes up to } \approx$$

ie i) $\sigma \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} \approx \begin{bmatrix} m(q_i) \\ k_j \end{bmatrix}$

ii) $\pi_1 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} \approx m(q_i)$

iii) $p \pi_2 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} \approx v_j$, for $i \in [n]$, $j \in [m]$

σ is the corpus fibration lifting

Characterize a domain of interest by an existential graph predicate on certain relationships between relevant subjects, to determine a subset of values $\hat{V}$ as follows:

Domain input data

Let domain keyword tokens be $v_1, \dots, v_L$, $L < m$

$$\text{ie } \{v_1, \dots, v_L\} \subset \{v_1, \dots, v_L, v_{L+1}, \dots, v_m\} \subset V$$

$$\hat{K} \triangleq \bigcup_{j \in [L]} p^{-1}(v_j) \subset K$$

$$\hat{V} \triangleq p(\hat{K}) \subset V$$

Note that $v_1, \dots, v_L \in \hat{V}$

Let $\hat{p} : \hat{K} \rightarrow \hat{V}$, then by definition
 $k \mapsto p(k)$

$$\begin{array}{ccc} \hat{K} & \xhookrightarrow{\quad} & K \\ \hat{p} \downarrow & & \downarrow p \\ \hat{V} & \xhookrightarrow[\quad]{\quad} & V \end{array} \quad \text{commutes.}$$

$$\begin{array}{ccccc}
 Q \otimes \hat{K} & \xrightarrow{1 \otimes l} & Q \otimes K & \xrightarrow{m \otimes 1} & Q' \otimes K \\
 1 \otimes \hat{p} \downarrow & & \hat{\sigma} \downarrow & \nearrow \sigma & \downarrow 1 \otimes p \\
 Q \otimes \hat{V} & \xrightarrow{1 \otimes l} & Q \otimes V & \xrightarrow{m \otimes 1} & Q' \otimes V
 \end{array}$$

$$\hat{\sigma} \triangleq \sigma \circ (1 \otimes l) : Q \otimes \hat{V} \rightarrow Q' \otimes K$$

is a domain specific fibration lifting

and by definition σ is compatible with $\hat{\sigma}$.

$$\text{Note that } \hat{\sigma} \begin{bmatrix} q_i \\ v_j \end{bmatrix} = \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix}, \quad i \in [n], j \in [l]$$

that is, domain specific attention is the corpus

trained fibration abductive attention restricted

to $\hat{V} \subset V$.

References

[1] 'Attention Entanglement'

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Transformer attention uses statistical correlation to update queries, and fibration abduction attention on top, takes into account key-value semantics, finding a key to explain the value returned. The latter has a quantum interpretation, indeed a fibration lifting is an entanglement. Entanglements (inner loop) can be refined using a Renyi α -entropy criterion, as part of a simultaneous query and key update (outer loop). Thus the value returned then reflects refinement of both query-key similarity and key-value semantics.

[2] 'On the Algebra of Logic: A Contribution to the Philosophy of Notation'

Charles Sanders Peirce, 1885

American Journal of Mathematics, Volume 7.